DEPI | Digital Egypt Pioneers Initiative

Track: Data Scientist - Capstone Project

Retail Intelligence & Market Basket Analysis Platform

An end-to-end business analytics decision platform

Presented by

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Milestone M1 - 16 October 2026 - Version 1.0

The Business Problem

Data-rich retailer, evidence-poor decisions

The retailer owns thousands of purchase records but cannot answer four practical questions:

Products

What do customers buy together?

Customers

Who are the most valuable customers?

Recommendations

What should we suggest to each customer?

Business

How do we raise average order value?

Result today

Decisions on shelf layout, product bundles and campaigns are taken without evidence.

Our answer: turn raw sales transactions into actionable, explainable recommendations.

Proposed Solution

Three connected insights, one dashboard

Customer Segmentation

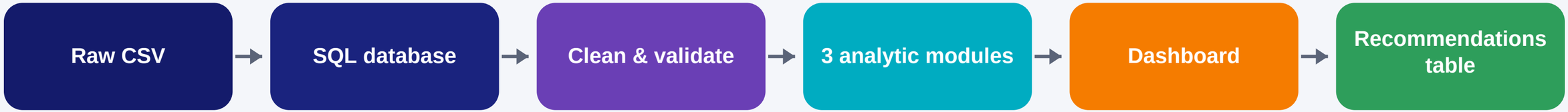
Unsupervised clustering on behaviour: recency, frequency, spend, discount reliance

Market Basket Analysis

FP-Growth rules with support, confidence and lift: what is bought together

Recommender System

Hybrid model: suggests the next products to add, compared to a popularity baseline



Every cleaning decision is logged, so each number can be traced back to the source.

The Data

Retail_Data from Kaggle (publisher: tawfiqakram)

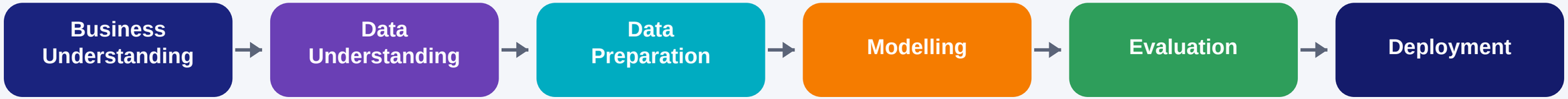
~0.8M	25	7,000	4
transaction lines	stores	customers	related tables

transactions	line id, invoice id, customer id, date, product, quantity, unit price, discount, store, payment method, channel
customers	customer id, city, age group, gender, join date, loyalty member, preferred payment
products	product id, name, category, brand, cost price, list price, perishable flag
stores	store id, name, city, area, region, size class, opening date

Privacy: anonymised identifiers only. Dataset licence to be confirmed and recorded in the README.

Methodology & Objectives

CRISP-DM in short weekly iterations



O1	Reproducible data pipeline + issues log	6 Nov
O2	Customer segmentation (Silhouette + bootstrap stability)	16 Nov
O3	Product associations with FP-Growth	16 Nov
O4	Recommender vs popularity baseline (time-based hold-out)	25 Nov
O5	Interactive dashboard answering all stakeholder questions	27 Nov
O6	Recommendations table, report, presentation, infographic	4 Dec

Project Plan

DEPI milestones and phases

16 Oct	6 Nov	30 Nov	4 Dec
M1	**M2**	**M3**	**M4**
Planning documents	Analysis & design, data pipeline ready	Implementation complete, source frozen	Final presentation, testing & reports

Done

Our Team

Five members, one role each, backups and peer-reviewed pull requests

Sherif Ayman

R1 Lead + Communication

Business case, KPIs, requirements, final report, presentation

Sama Dawood

R2 Data Engineer

SQL database, loading scripts, ERD, reproducible pipeline

Ali Mohamed

R3 Cleaning + EDA

Profiling, cleaning rules, issues log, EDA notebook

Menna Magdy

R4 ML + Evaluation

Features, clustering, association rules, recommender

Rawan Bassem

R5 Dashboard + Deployment

Dashboard, wireframes, number checks, optional API

Team Leader: Sherif Ayman

Working agreements: weekly 45-min meeting, protected main branch, every pull request reviewed, cross-training so anyone can explain the whole project.

Success Metrics

18 KPIs across delivery, data quality, models and system

100%	**>=99%**	**0**	**>=0.80**
Milestones on time (4 of 4)	Key completeness after cleaning	Duplicate rows after cleaning	Segment stability (Adjusted Rand Index)
>=1.25x	**>=2x**	**<=5 s**	**<=1 s**
Recommender Hit@10 vs popularity baseline	Recommender Hit@1 vs baseline	Dashboard page load time	Top-5 recommendation response

>=90% test pass rate **>=5 actionable recommendations** **Pipeline reproducible from README**

Analytical targets are relative to baselines, so they stay meaningful if the dataset changes.

Risk Assessment

Probability x impact, reviewed weekly when score >= 6

9	**R01 Cleaning takes longer than planned** Mitigation: Profile in week 1; model on a sample first
6	**R03 Data leakage inflates results** Mitigation: Time-based split; peer review of every feature
6	**R07 Scope creep** Mitigation: Frozen scope; optional items only after M3
6	**R09 Dashboard differs from SQL results** Mitigation: Single KPI definition; checks against SQL
6	**R04 Limited tool experience** Mitigation: Learning plan per role; pair programming; backups
6	**R11 Deadline pressure before M2, M3, M4** Mitigation: Finish one week before each DEPI date

Also tracked: dataset licence, merge conflicts, large files, misinterpretation, availability, personal data in repo.

Literature Review

What we adopt from existing work, and how we differ

What we adopt

- FP-Growth over Apriori for speed (Han, Pei & Yin, 2000)
- Use lift, not only confidence, to avoid popularity bias
- Go beyond RFM: add discount use, basket size, visit gaps, tenure
- Time-based evaluation, as in the Instacart competition
- Item-based collaborative filtering + gradient-boosting re-ranker
- Validate on Online Retail II (UCI) as a second dataset

How we differ

- Three connected modules on one cleaned dataset
- Data cleaning is a deliverable, with a logged before/after
- Ends in a recommendations table: priority, owner, metric
- Every recommendation carries a plain-language reason
- Fully open-source and reproducible, unlike managed services such as Amazon Personalize

Sources: 15 references reviewed (papers, Kaggle, UCI, AWS documentation).

Requirements

Gathered with stakeholders, prioritised with MoSCoW

8	15	8	26	17
Stakeholder groups	User stories	Use cases	Functional requirements	Non-functional requirements

Stakeholders

- Store / Commercial Manager
- Marketing and CRM teams
- Merchandising and Operations
- Data / IT team
- Supervisor and project team

Functional

- Cleaning pipeline + issues log
- KPIs, trends, rankings
- Clustering and segment profiles
- FP-Growth rules with lift
- Top-N basket recommender
- Dashboard + recommendations table

Non-functional

- Dashboard render <= 5 s
- Recommendation <= 1 s
- Pipeline <= 15 min on 8 GB RAM
- Anonymised data, no secrets in repo
- Reproducible, explainable, portable

Deliverables & Next Steps

Deliverables

- Planning documents on GitHub (M1, done)
- System analysis & design: ERD, DFD, UML, wireframes
- SQL scripts, notebooks, source code and README
- Interactive dashboard
- Recommendations table and final report
- User manual, presentation and infographic

Next Steps

- Confirm dataset licence and dashboard tool with supervisor
- Finish system analysis & design by 6 Nov
- Build and validate the data pipeline
- Segmentation, market basket and recommender
- Dashboard, testing and final delivery by 4 Dec

Thank you

Sherif Ayman - Sama Dawood - Ali Mohamed - Menna Magdy - Rawan Bassem